

MATHEMATICS

Modern Algebra I

Excercises 4. (From the Prelims!)

- (1) Show that there are no simple groups of order 616, 132, 1365, 245, 520, 156.
- (2) Show that a) any group of order 616 is solvable, b) any group of order 105 has a normal cyclic group of order 35, c) any group of order 289 is abelian.
- (3) If G is simple of order 60, then G is isomorphic to A_5 .
- (4) If G is finite simple with exactly 5 Sylow p -subgroups, for some prime p . Prove that G is isomorphic to A_5 .
- (5) Suppose that $|G| = p^n$, p is prime and $n > 1$. a) Prove that the group of all automorphisms of G is a finite group A with order divisible by p . b) Show that the center of G is nontrivial, c) if G acts on a set S consisting of r elements, show that the number of points fixed by G is congruent to r modulo p .
- (6) Let G be a non-abelian group of order p^3 , with p prime and Z its center. Show that $G/Z \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$.
- (7) Let G be a non-abelian group of order p^3 , with p prime, Z its center and G' its commutator subgroup. Show that $Z = G'$.
- (8) Let G be a finite abelian group, n an integer. Show that the map that sends g to ng is a group homomorphism and it is an isomorphism if and only if n is relatively prime to $|G|$.
- (9) Let G be a group of order 55 acting on a set S consisting of 24 elements. Show that there is at least one element fixed by every element of G .
- (10) Let G be a finite group such that for all positive integers n , there are at most 2 subgroups of order n in G . Prove that G is cyclic.
- (11) Show that the group generated by x, y with relations $x^2 = 1, y^2 = 1, xyx = yxy$ has 6 elements.